E-stop: 1 NC out, 2 GND, 3 NC return, 4 GND, 5–6 reset NO, 7 GND, 8 button NO
Buck board pigtail: 1–2 +VIN, 3–4 GND, 5 ON/OFF (low = on), 6 GND, 7 +3V3_PI, 8 SDA, 9 GND, 10 SCL, 11 SP_EN, 12 GND
HS50 B (−X): 1 VBUS
R leg: 14 AWG pigtail pair, 1 +, 2 return
2mR 2512 CSS2H−2512K−2L00F 5W
TPS48111LQDGXRQ1
IPB017N10N5LFATMA1
TPS3431SDRBR
MMBT3904

1. Battery in and E–stop bank B (LM5069–1, two FETs). Bank A, in series ahead of it, is block 7

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97–123 A (VCL 48.5–61.5 mV), breaker 160–260 A (VCB 80–130 mV). Above peak draw and the BMS DOC (70 A or more). 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10–22 uA, 100–220 V/s, 0.25–0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 652 W at 54.6 V, 480–810 W with tolerance: RPWR = 1.3e5 x Rs x (P − 1.18 mV x VDS / Rs), SNVS452G eq 10, VSNS 6 mV over the 5 mV floor. Timer 4.7 nF: 0.22 ms typ, 0.15–0.39 ms.
Fault check: 810 W is 14.8 A at 54.6 V for up to 0.39 ms. Reviewers read about 16 A at 0.82 ms, 70 C, one FET; TI scales as t^0.346, so about 20.7 A at 0.39 ms: 1.4 x. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A, the same again in bank A. UVLO low sinks the gate at 2 mA: about 170 us off.
The 2 mA also pulls VBUS down through U1's OUT–GATE zener, against the 54 uA of its internal 1 MOhm SENSE–OUT path, so VBUS reads near 0 V with the banks open.
Body diodes of both banks pass regen back to the pack with the banks open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069–1 latches on a fault. Clear: E–stop cycle, then reset.
BAT spike: the absorber in blocks 10 and 10d (8 x 47 uF) takes the lead current when a bank opens fast, 5.0SMDJ58A is the backstop. FETs and U1 are 100 V.
Polarity (rev B5): one soldered 8 AWG pigtail pair, side by side, to an XT90 that only mates the pack's XT90S one way (Joey: the pack run is the heaviest connector run in the robot). Nothing on the main path blocks a reversed pack: both banks' body diodes would conduct and D1 becomes the crowbar, so the keyed plug and the bring–up polarity check carry it.

Block 10. Absorber: a bank trip into a bus short cuts 350–890 A. tools/absorber_sim.py (pack 54.6 V, no pack yet), worst corners: 0.05 Ohm pulse loop (cell AC resistance), 0.5–1 uH, D1 at 71.2 V, C −20 %, trip 8.8–17.5 us. Eight 47 uF 80 V hybrids (EEH–ZC1K470P, the bus–cap part) give BAT 75 V or less (0.05 Ohm loop): 6 % under rating, about 18 % under surge (rev B5: the APZ 100 V line is obsolete). Bring–up step 1 is a gate: a pulse loop under 0.05 Ohm means rerun absorber_sim before motors. Measure lead L and pulse R at bring–up. 100k bleeds it. BMS close drives about 300 A into three cans for a few us: bench it on the JK.
U2 feed: F1, 22R, ES1D in series, then 1 uF. A shorted U2 puts the pack on VCTL: 22R holds it to about 1.6 A, so SMAJ15A clamps under 19.2 V (Q6 gate 20 V) and F1 (50 A breaking) opens. The series diode keeps C5 from back–driving U2 when BAT collapses, and 22R keeps C39's inrush far under F1's I2t.
Absorber close option (M3), DNP: if the BMS trips when it closes onto the absorber, remove the 0R and fit 2R + the delayed NMOS + its RC: 25 A at close, clamp unchanged (absorber_sim.py).
RUN_A and RUN_B diodes: a U1 pin 2–3 bridge or an EN fault can't drive RUN. FLT reaches UVLO through BAT46W + 10k (2010) with a 12 V zener on FLT_A: FLT low holds UVLO near 1.9 V; a bridged UVLO puts at most 5.5 mA into FLT and 12.7 V on it (15 V recommended).
Converter: F2 20 A on the board end protects the pigtail (U7 already limits at 9.7–12.3 A; F2 backs up a shorted Q7). The DVE has no reverse protection, so the keyed connector carries polarity. U7 PGD high (good) turns on the 2N7002, which pulls ON/OFF to −Vin through 220R: on (0.45 V at 2 mA). 220R limits a pin 5–11 short to 55 mA. 470k and the 12 V zener bias the gate from BRK_IN, so it doesn't depend on VCTL.
1 uF on RST_SRC (M8): a reset held at power–up no longer arms the latch (tools/latch_sim.py, 36 cases). Reset interlock (M1): Q10 shorts RST_SRC until MID is over about 11 V (10 V zener, Q11), and Q14 shorts it while VBUS is over 6–8.5 V (5.1 V zener, hFE spread). Operator rule: wait for VBUS under 3 V. Reset needs MID healthy and the bus down, so a shorted B can't be hard–closed onto a charged bus. After a stop, wait for the drives' caps to bleed. It loads MID by about 30 uA. 33k on U17 VBUS (M13). 22 nF SRC to GND, TI's recommendation for U8. BAT46W (M4). 330 uF on +12V: spare–output shorts and servo back–feed (M9) land in it and the compute loads.

2. Control rail and E–stop latch (no logic IC)

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.
Fed through F1 with its own 1 uF, clamped by a 15 V TVS, reverse diode OUT to IN: block 10.

Seal–in latch, ngspice checked at hFE 80: stays off at power–up even with reset held, reset closes it, a 50 us contact bounce rides through, loop open drops it in about 100 us, re–closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V. UVLO and EN_A each take RUN through their own diode (block 10), so a fault on either pin cannot drive RUN. UVLO 3.8 V; needs about 4.8 V on RUN_B rising. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control–rated NC contact. No 12 V on the connector without a series R. GND between loop out and return, and between return and reset: a short there opens the loop or does nothing.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts.
Button HB2–BS545, 22 mm mushroom, twist release, 1NC + 1NO, 660 V AC 10 A, control–rated only: NC in this loop, NO to the Pi (block 9). RUN also enables bank A (block 7) and SPARE_SW (block 8).

3. Braking clamp: 56.6–56.8 V on, 55.8–55.9 V off (typ), 57.5 V worst case, into 2 x HS50 20R in parallel

Two Arcol HS50 20R J in parallel, 10 Ohm total, one on each gusset (+X: J3/J4; −X: J25/J26, block 10c; place all four on the board's top edge, +X half, because the −X leads cross under the board, about 150 mm longer), on the 5 mm countersunk Al plates with thermal compound. Each [illegible] 322 W total, 2.84 A and 161 W per resistor. MBRS3100 carries the lead and winding inductance at turn–off, so lead length sets EMI, not FET stress. Each is rated 50 W only on Arcol's 535 cm2 plate, so it works on burst energy: 32 g each, about 29 J/K, so a 3 kJ burst is about +50 C per body. Software caps braking at 50 W average over 60 s and 3 kJ per burst, total. Thermostats: one 67L120 under each body, both in series in the E–stop loop (rev B5), so an open resistor can't leave t
Self–powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the resistor's thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

235 uF, 80 V: 5 x Panasonic EEH–ZC1K470P hybrid, 36 mOhm, 10 x 10.2 mm, active line (rev B5: the APZ 68 uF is obsolete); the fifth is in block 10d. The drives have their own bulk. VBUS peaks about 58 V: 27 % margin. Regen overshoot about 0.8 V before the clamp acts.

5. 12 V converter and 12 V distribution: on the buck board (buck12), on the J24 pigtail (block 10)

Split S1 (2026–11–09): the 12 V converter and everything on +12 V (the 12 V meter, the SPARE_SW and SPARE_ON eFuses, the fused Jetson, Pi–buck and fan outputs, the bulk cap and TVS) are on the buck board. +12 V never comes back here.
J24 carries the 54 V feed and ON/OFF (as before), the Pi's +3V3 and I2C for the buck board's two meters, and SP_EN (block 5c).
INA228 addresses, no strap on SDA: L arm 0x40, R arm 0x41, aux 0x43, L leg 0x44, R leg 0x45, SPARE_SW 0x47 (buck board), waist 0x4C, 12 V 0x4D (buck board), MID 0x4F. Powered by the Pi's 3.3 V; Pi pull–ups on I2C.

5a. Brick soft start (ahead of the E–stop switch)

Same controller as the bank, so one part family. −2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power–limit start, no dV/dt network: 70 W limit (Rs 5 mOhm, same equation as U1), about 1.3 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 100 nF: 4.7 ms typ, 3.1–8 ms. Limit 9.7–12.3 A clears the converter's worst draw, about 4.2 A at 36 V. It charges about 48 uF at the converter end plus the DVE's unpublished internal capacitance at the low power limit.
Fault check: 87 W is 1.6 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

6. Limb ports: 2 mOhm shunt, own breaker (block 6a), meter, soldered pigtail pair. Pigtail fuses inline.

[illegible]
Each limb's + and return leave on one soldered pair, twisted from the board edge. No GND studs.
Aux and legs DNP: fit the same parts when they arrive (legs on rev C).
U12 reads the R arm port input, so it shows VBUS with every port off; the other meters read their port outputs.

6a. Limb port breakers: TPS48111–Q1 + one FET per port, latch–off, enabled by RUN

Why: a hard limb short used to latch the whole bus (U1 trips in under 0.4 ms, long before a 20 A fuse melts). Each port now opens itself first.
Short circuit: RISCP 4.64k on 2 mOhm = 35–45 A (15.6 uA x (R + 464)); the shunt is a 2–terminal 5 W metal strip on the split–pad Kelvin land (each end bridges its current and sense pads), 1.8 W at the 30 A trip level, PD low within 1.6 us, then 22R to the gate: Miller about 0.16 us, so the drain–side di/dt stays a few A/ns against the 2.2 uF and the bus caps. tools/port_sim.py: for 50 nH to 1 uH of wiring, 0.5–1 uH pack lead, 0–50 A on the other limbs, RS1 peaks at 130 A or less against U8 158 A and U1 160 A minimum: the port always opens first. A short at the board pads (50 nH) puts about 1.1–1.4 kA through the FET for under 2 us (5–14 mJ; IPB017: 1024 A pulse, 979 mJ avalanche), which is why the port uses the bank FET. A short in the harness (200 nH or more) is 400–540 A.
Overload: RIWRN 20k = 30 A (11.9 x RSET / (RSNS x IOC)), CTMR 1 uF with 100k across it: about 16 ms, then latch–off. The 20 A pigtail fuse stays as wire protection and as backup for a shorted FET.
Latch clears only when EN drops: an E–stop cycle, then reset. FLT_I of every port pulls PORT_FLT_N (Pi header): the Pi names the port whose output is low.
Start: EN and INP come from RUN through one diode (a pin fault can't drive RUN). The bus starts from 0 under U1, so the port wakes at VS 3 V with BST up (0.22 uF, about 20 ms): the gate rises through 100k into 100 nF (diode–coupled, 10M bleed, so PD never sees it), about 1 kV/s at the output: 20 mF of drive capacitance draws about 20 A, under the 30 A overload level. After that the drive caps follow the bus.
Off (E–stop, trip): the body diode still returns regen from the limb to the bus and the clamp. SK810L on each output carries the wiring current when the FET opens. 15 V zener gate to source.

7. E–stop bank A: second series bank, own driver, ahead of bank B

Own driver: a FET usually dies shorted gate to drain, which would pull a shared gate net to the pack and hold the other bank on. RUN is the only shared signal, and each bank takes it through its own diode.
Both banks face the same way, drains toward the pack, so regen stays open. Bank B keeps U1, the shunt and the soft start.
Close order: U8 FLT holds U1 UVLO low (about 1.9 V, through BAT46W + 10k) until BST is up (about 57 ms typ, 35–115 ms with 2.2 uF), so A closes first by logic, not by timing. The diode only lets FLT pull down: a failed–high FLT can't enable B. A closes onto MID only. FLT also blocks B if U8 has no charge pump or has tripped.
Off edge: EN falls first, 4.6 us typ, 6.5 us max, then G1PD pulls 390 nC through 22R: about 2–3 us. On a bus short U1's breaker can open B first, so the spike can land on MID with A on: the absorber (block 10) bounds BAT, and so MID, and U17 has its own clamp.
Short–circuit comparator on RS1: 40.2k gives 99 mV, 158–242 A, above U1's limit. CS+/CS− through 0R with a DNP cap: TI's RC placeholder against nuisance trips. TMR 100k only (C29 DNP): fastest response, 4.3 us, still latch–off, cleared by an E–stop cycle. It covers reset with B shorted: A opens, and FLT holds B off.
Driver off holds gate to source with no path to GND, so MID can be biased: 47k + 47k, 0.47 x pack nominal, 0.6 mA, 16 mW each. Hot FET leakage (200 uA per bank at 125 C) moves MID by up to about 4 V. U17 reads MID through 33k with a 68 V zener at the pin (85 V max): calibrate the 3 % gain.
Self–test before reset, pack 36–54.6 V, BMS telemetry for pack voltage: MID 0.33–0.60 x pack and VBUS under 3 V is healthy (U11 reads V_L1: use the highest limb meter). MID within 2 V of pack: A shorted. MID under 3 V: B shorted. VBUS high: both shorted, or the bus has not bled down yet, about 11 s on board caps, longer with drives.
U17 reads MID on its VBUS pin, and clamp duty on IN+ (block 10d). VBUS alone cannot see one shorted bank, which is why MID is read.

8. Spare 12 V outputs: on the buck board (split S1). SPARE_SW is enabled from RUN through block 5c

9. E–stop pressed: button NO contact to the Pi

Low means pressed. Pi GPIO input, no Pi pull–up needed. 3.3 V through 3.3k gives the contact 1 mA of wetting.
A harness short to the 54.6 V bus: the BAT46W blocks the pull–up (block 10), and 100k leaves 0.5 mA in the clamp pair. Open contact reads about 3.0 V.
Information only: the NC loop does the stopping.

10. Rev B2 protection parts (refs appended so earlier refs keep their numbers)

10b. Absorber close option (M3): 0R fitted, the rest DNP

10c. Rev B4 review parts

Rev B4 review: two more absorber cans (three in all). U2 feed 22R and series ES1D. FLT diode BAT46W (pull–down only). 68 V zener on U17 VBUS. 220R in Q12's drain. U8 sense RC placeholder: 0R fitted, 1 nF DNP. Schottky on SPARE_ON (and D5 on SPARE_SW). VBUS bleeder footprint, DNP: fit it only if the drives don't bleed the bus. Fuses on the Jetson, Pi buck and fan outputs (the DVE limits at about 30 A). SMBJ15A on +12V: servo back–feed clamp, 16.7 V min breakdown, so it never conducts at 12 V.

10d. Rev B5 parts

2a. Pi heartbeat watchdog: no heartbeat, no latch

The Pi toggles WD_FEED from its control loop (software, not a PWM peripheral) every 50–100 ms; the watchdog wants a falling edge at WDI at least every 270 ms (CWD 3.3 nF C0G: 77.4 x C + 55 ms = 310 ms typ, 270–355 ms with the +−9.5 % timer and 5 % cap band). Miss it and WDO goes low for 200 ms: the NPN in Q5's emitter turns off, the seal–in breaks and RUN falls like a loop open. The latch stays off after WDO releases: only a reset press re–arms it.
Fail–safe: the NPN only conducts while the watchdog's own rail (3.3 V zener from VCTL, not the Pi) feeds its base through 22k and WDO is released. Pi dead, unpowered, rebooting, hung or its pin floating (100k pull–down): no edges, no latch. Watchdog rail lost: no base drive, no latch. It sits in series, so it can only drop the latch, never hold it (a shorted NPN just removes the watchdog).
While the Pi stays dead the watchdog repeats 310 ms released / 200 ms low; a reset press in a released window closes the latch until the next low, under 0.36 s, with no drive commands (the Pi is dead). The first 0.3 s after VCTL comes up the watchdog is released before its first timeout.
Software requirement: drive CanTimeout set and read back before every enable.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/power–board/REV_B_PART
marengo
Sheet: /
File: marengo_pdu_revb.kicad_sch
Title: marengo power board rev B
Size: A0 | Date: 2026–11–06 | **Rev: B–wip**
KiCad E.D.A. 10.0.8 | Id: 1/1